

Kendall Thomas

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Education

- **University of North Carolina at Chapel Hill** **Est. May 2027**
 - Ph.D. Statistics and Operations Research (Advisor: Jan Hannig)
 - Dissertation Title: Statistical Frontiers in Elite Athlete PerformanceChapel Hill, NC
- **University of North Carolina at Chapel Hill** **Dec. 2025**
 - M.S. Statistics, Analytics, and Data ScienceChapel Hill, NC
- **Davidson College** **May 2019**
 - B.S. Mathematics, B.S. Computer Science
 - Magna Cum LaudeDavidson, NC

Professional Experience

- **Graduate Teaching Fellow** **Jan. 2026 – Present**
 - *University of North Carolina*
 - *Instructor of Record: STOR 320, STOR 455***May 2024 – July 2025**
Chapel Hill, NC
- **Graduate Research Assistant** **Aug. 2025 – Dec. 2025**
 - *University of North Carolina*Chapel Hill, NC
- **Graduate Teaching Assistant / Lab Instructor** **Aug. 2022 – May 2024**
 - *University of North Carolina*Chapel Hill, NC
- **Software Engineer (promoted to Software Engineer II)** **Aug. 2019 – Apr. 2022**
 - *Microsoft, Azure Data*Redmond, WA
- **Research Assistant** **Aug. 2016 – May 2019**
 - *Dr. Tim Chartier, Davidson College*Davidson, NC
- **Assistant Analyst** **Jan. 2018 – Dec. 2019**
 - *John Tobias, ESPN and FOX Sports*Remote
- **Advanced Analytics and Machine Learning Intern** **May 2018 – Aug. 2018**
 - *Nike*Beaverton, OR
- **Research Assistant** **Feb. 2017 – Nov. 2017**
 - *Athlete Intelligence*Seattle, WA / Remote

Publications

Under Review / Submitted

- 1) Thomas, K. L., & Hannig, J. (2025). *Movement dynamics in elite female soccer athletes: The Quantile Cube Approach*. Manuscript in revision for the *Journal of Quantitative Analysis in Sports*. Preprint available at <https://arxiv.org/abs/2503.11815>.¹

Teaching

Curriculum Developed

- “Introduction to Data Science in Python”
 - 1) Led a curriculum redesign introducing a Python-based section of STOR 320 *Methods and Models of Data Science* (first offered Summer 2024; renamed from Introduction to Data Science in Fall 2025) alongside the existing R-based version.
 - 2) Developed new lectures and Jupyter-based lab modules emphasizing data acquisition, cleaning, visualization, modeling, and communication in Python.
 - 3) The Python section is now offered each semester, providing students an alternative track aligned with modern data science tools and reproducible workflows.

Courses Taught (Instructor of Record)

- **Scheduled: Summer Session I & II 2026**
STOR 455 (Methods of Data Analysis)
- **Spring 2026** 42 students
 - 1) STOR 320 (Methods and Models of Data Science)
- **Summer Session II 2025** 15 students
 - 2) STOR 455 (Methods of Data Analysis)
- **Spring 2025** 94 students
 - 3) STOR 320 (Introduction to Data Science)
- **Fall 2024** 101 students
 - 4) STOR 455 (Methods of Data Analysis)
- **Summer Session II 2024** 22 students
 - 5) STOR 320 (Introduction to Data Science)

Awards and Honors

Academic

- **Druscilla French Graduate Student Excellence Fund Award** Aug. 2025 – May 2026
 - *University of North Carolina at Chapel Hill*

¹ Related Paper, Presentations, and Awards

- **Walter Deemer Excellence in Teaching Award** **Dec. 2024**
 - *University of North Carolina at Chapel Hill Department of Statistics and Operations Research*
- **Excellence in Teaching Assistance and Instruction Award** **Dec. 2023**
 - *University of North Carolina at Chapel Hill Department of Statistics and Operations Research*

Conference & Research Competitions

- **First Place Poster Award¹** **Sept. 2025**
 - *New England Symposium on Statistics in Sports (NESSIS)* Boston, MA
- **First Place Poster Award¹** **Nov. 2024**
 - *Carnegie Mellon Sports Analytics Conference* Pittsburgh, PA
- **Hackathon Finalist (Professional Division)** **Mar. 2020**
 - *MIT Sloan Sports Analytics Conference* Boston, MA
- **Grace Hopper Celebration Research Scholar** **Oct. 2017**
 - *Computing Research Association – Women* Orlando, FL

Industry & Technical Hackathons

- **Winner - Greatest Business Value** **Fall 2021**
 - *Microsoft Azure Supportability Hackathon*
- **Winner – Most Impactful & Greatest Business Value** **Spring 2021**
 - *Microsoft Azure Supportability Hackathon*
- **Winner – Best Hack (Audience Vote)** **Fall 2019**
 - *Microsoft Azure Data Hackathon*

Leadership, Service, & Outreach

- **Graduate Student Liaison** **Jan. 2024 – Present**
 - *University of North Carolina at Chapel Hill Department of Statistics and Operations Research*
- **Graduate Student Mentor** **Aug. 2023 – Present**
 - *University of North Carolina at Chapel Hill Department of Statistics and Operations Research*
- **Sports Analytics Intelligence Lab (SAIL) - Principal** **May 2025 – Present**
- **Sports Analytics Intelligence Lab (SAIL) - Graduate Student Lead** **Jan. 2023 – Present**
 - *University of North Carolina at Chapel Hill Department of Statistics and Operations Research*

¹ Related Paper, Presentations, and Awards

- **Varsity Athlete NCAA Division I Women's Soccer Team** **July 2015 – May 2019**
 - *Davidson College*
- **Analyst & Team Manager NCAA Division I Men's Basketball Team** **Jan. 2018 – May 2019**
 - *Davidson College*
- **'Cats Stats Lead Analyst** **Jan. 2016 – May 2019**
 - *Davidson College*
- **Females in Computer Science and Information Technology President** **Aug. 2017 – May 2019**
 - *Davidson College*

Presentations

Invited Talks

- **“Analyzing Movement Dynamics in Elite Female Soccer Athletes”¹** **Scheduled Apr. 2026**
 - Related Research Authors: Thomas, K. L., & Hannig, J. **Mansfield, CT**
 - *Connecticut Sports Analytics Symposium*
- **“Analyzing Movement Dynamics in Elite Female Soccer Athletes”¹** **Jan. 2026**
 - Related Research Authors: Thomas, K. L., & Hannig, J. **Houston, TX**
 - *American Soccer Insights Summit*
- **“Beyond a Bit Fit”** **Apr. 2019**
 - Related Research Authors: Thomas, K. L. & Chartier, T. **Greenville, SC**
 - *Carolina Sports Analytics Meeting*

Conference Presentations

- **“Statistical Methods for Estimating Player Performance Across Ages and Leagues”** **Scheduled Aug. 2026**
 - Topic-Contributed Session, *Joint Statistical Meetings (JSM)* **Boston, MA**
- **“Analyzing Movement Dynamics in Elite Female Soccer Athletes”¹** **Nov. 2024**
 - Related Research Authors: Thomas, K. L., & Hannig, J. **Virtual**
 - *Midwest Sports Analytics Meeting*
- **“Beyond a Bit Fit”** **Nov. 2019**
 - Related Research Authors: Thomas, K. L. & Chartier, T. **Pella, IA**
 - *Midwest Sports Analytics Meeting*

¹ Related Paper, Presentations, and Awards

Conference Posters

- “Analyzing Movement Dynamics in Elite Female Soccer Athletes”¹ **Feb. 2026**
○ Thomas, K. L., & Hannig, J. Chapel Hill, NC
○ *CoreAI UNC Research Showcase*
- “Analyzing Movement Dynamics in Elite Female Soccer Athletes”¹ **Sept. 2025**
○ Thomas, K. L., & Hannig, J. Boston, MA
○ *New England Symposium on Statistics in Sports*
- “Analyzing Movement Dynamics in Elite Female Soccer Athletes”¹ **Nov. 2024**
○ Thomas, K. L., & Hannig, J. Pittsburgh, PA
○ *Carnegie Mellon Sports Analytics Conference*
- “Professional vs. Elite Female Soccer Athletes: Physical and Technical Performance Characteristics” **Jul. 2024**
Baltimore, MD
○ Cantú, E. I., Moore, S. R., Thomas, K. L. & Smith-Ryan, A. E., FNSCA.
○ *National Strength and Conditioning Association National Conference*

Professional Memberships

- American Statistical Association (ASA)
- Association for Women in Mathematics (AWM)
- Phi Beta Kappa Society
- Omicron Delta Kappa: The National Leadership Honor Society

Technical Skills

Programming Languages

- Advanced: Python, R
- Intermediate: Java, HTML, CSS, SQL, Kusto
- Basic: C++, C#, C, JavaScript, p5.js, d3.js

Software and Tools

- Advanced: LaTeX, R Studio, Microsoft Office, Tableau, Jupyter Notebooks
- Intermediate: SQL Server Management Studio, Azure Data Studio
- Basic: Mathematica, ArcGIS, MATLAB

Libraries and Frameworks

- Data Manipulation & Analysis: pandas, NumPy, SciPy
- Data Visualization: ggplot, matplotlib, Plotly, seaborn
- Interactive Applications: Shiny (R and Python)
- Machine Learning / Data Science: scikit-learn, TensorFlow, PyTorch
- Web Scraping & Automation: BeautifulSoup, requests, Selenium
- Version Control / Workflow: Git / GitHub